Developing RSE skills at the novice level

Student RSE model for research support, experiential learning, and technical workforce development

Paula Lackie, Oliver Hall, Tim Dennis, Leigh Phan
Carleton College DataSquad | UCLA Library Data Science Center DataSquad

Paid students contribute to research and build RSE skills while they do it.

Campuses need more data and software help than staff can provide. Courses leave gaps, and so does the code researchers write. DataSquads fill those gaps.



A paid job that teaches

Many students never hear of these opportunities, especially students from underrepresented groups. Library or department staff supervise students. Students help researchers with their data, from cleaning to reproducibility.

Who is on a DataSquad

- **Project manager** triages requests
- **Data analysts and developers** do the hands-on work with data and code
- **Technical writers** document projects
- **Staff** supervise, mentor, fund, and sustain the DataSquad

Limits we keep

We help with a researcher's data and code. We don't change their research questions, tutor their students, or share anything a researcher tells us in confidence.

Pace of work

Schedules, the academic calendar, onboarding, review, and graduation handoffs set the pace of work. More students can take more varied projects but need more oversight. Handbooks, templates, and staff stay when students graduate.

What the work looks like

CARLETON Election-news scraper. Seven students built it in 2022. Two more extended it from 2023 to 2024 for a political science project that involved over 500,000 transcripts.



Each dot is one student. Blue built it, orange extended it.

UCLA Police helicopter flight data. Two students brought software skills to a research lab's R code. They refactored it with staff, and the code went through code review. Loading the data fell from about 318 minutes to about 2.



LAPD helicopter, for illustration. Photo: Chris Yarzab, CC BY 2.0, via Wikimedia Commons. Cropped and resized.

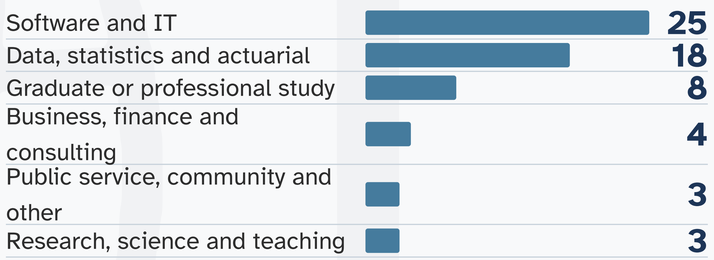
Time to load the data



Research software habits, when the project needs them. Run instructions, a dependency list, one test, and a citation file. We do not treat them as a checklist. We plan to track whether researchers keep them. Two student-built projects have citation files and DOIs that credit their developers: wbcheck (doi:10.5281/zenodo.23329531) and the UC OSPO Education site (doi:10.5281/zenodo.23330022).

Where students go next

Current job category of former student employees, from public profiles in 2026.



61 of 132 alumni coded so far: UCLA 19 of 19, Carleton 42 of 113. Carleton is only partly coded, so this is not a sample. Descriptive, not a placement rate.

What adopting takes

- Student positions and a wage budget
- Supervisors with time to review and coach
- Onboarding and a home for documentation
- A plan for turnover

Start a DataSquad

Get the starter kit for roles, onboarding, confidentiality, and request tracking. Students learn software sharing from our open lesson, **Sharing Research Software**. Supervise students? Compare onboarding, review, and graduation handoffs with us.

datasquad.info/join